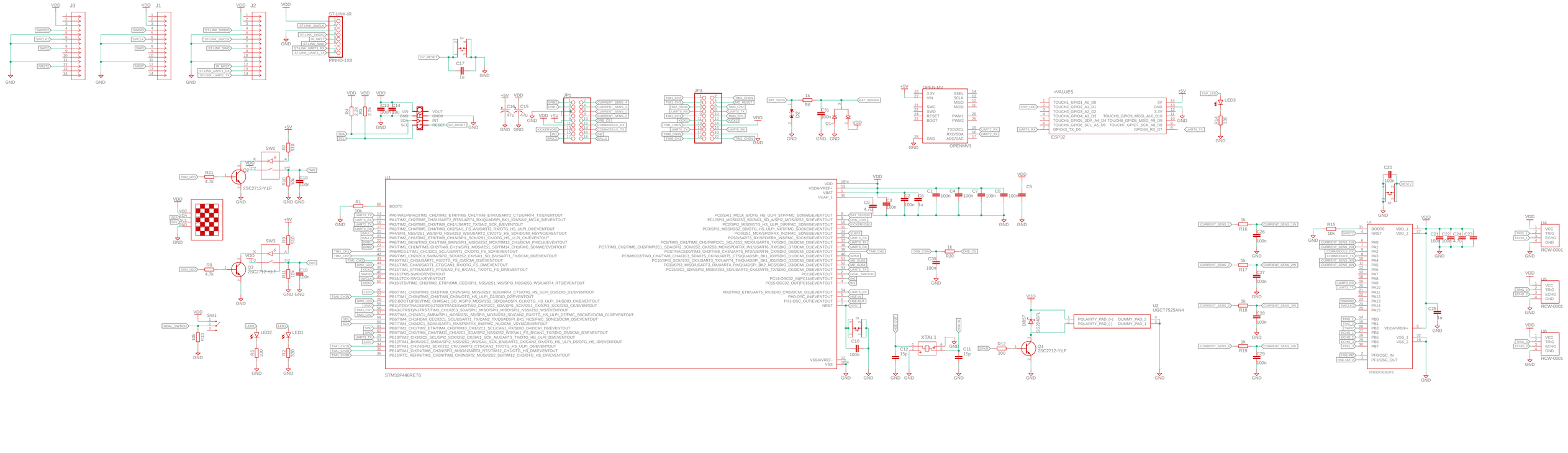

J3
J1
J2
ST-LINK-IR
PINHD-1X8
SWDIO
SWCLK
SWO
NRST
ST_LINK_SWDIO
ST_LINK_SWCLK
ST_LINK_SWO
ST_LINK_NRST
ST_LINK_UART1_RX
ST_LINK_UART1_TX
IR_NRST
S4
GY_RESET
C17
1u
C13
C14
VIN
GND
VOUT
GND
INT
SDA
SCL
RESET
R4
R5
2.2K
GY_RESET
SDA
SCL
C16
C15
4.7u
JP1
DIR0
DIR1
KICKER(D0)
BALL1
CURRENT_SENS_4
CURRENT_SENS_3
CURRENT_SENS_2
MB_CS
COMMOCULE_RX
COMMOCULE_TX
BALL2
JP3
TIM1_CH1
TIM1_CH2
BAT_SENS
UART3_RX
PWC1
TIM1_CH3N
TIM1_CH3
PWC2
UART3_TX
TIM1_CH3N
TIM1_CH2
TIM1_CH4
BAT_SENSN
MS_RESET
TIM5_CH2
UART5_TX
TIM1_CH1
UART5_RX
BAT_SENS
R6
1k
Q1
100nF
D1
D2
OPEN-MV
3.3V
VIN
SWC
SWD
RESET
BOOT
SSEL
SCLK
MISO
MOSI
PWM1
PWM2
TXD/SCL
RXD/SDA
ADC/DAC
5V
GND
3.3V
UART2_RX
UART2_TX
OPENMV3
>VALUES
TOUCH1_GPIO1_A0_D0
TOUCH2_GPIO2_A1_D1
TOUCH3_GPIO3_A2_D2
TOUCH4_GPIO4_A3_D3
TOUCH5_GPIO5_SDA_A4_D4
TOUCH6_GPIO6_SCL_A5_D5
GPIO43_TX_D6
TOUCH9_GPIO9_MOSI_A10_D10
TOUCH8_GPIO8_MISO_A9_D9
TOUCH7_GPIO7_SCK_A8_D8
GPIO44_RX_D7
ESP32
ESP_LED
UART4_RX
UART4_TX
LED3
R14
330
SW2
SW3
SW_LED
R21
4.7k
R8
R10
R11
R7
R9
510
10k
C19
C18
100n
Q2
Q3
2SC2712-YLF
VCC
SDA
SCL
GND
SW1
GOAL_SWITCH
R13
LED2
LED1
R5
R3
330
U1
STM32F446RET6
BOOT0
PA0-WKUP(PA0)/TIM2_CH1/TIM2_ETR/TIM5_CH1/TIM8_ETR/USART2_CTS/UART4_TX/EVENTOUT
PA1/TIM2_CH2/TIM5_CH2/USART2_RTS/UART4_RX/QUADSPI_BK1_IO3/SAI2_MCLK_B/EVENTOUT
PA2/TIM2_CH3/TIM5_CH3/TIM9_CH1/USART2_TX/SAI2_SCK_B/EVENTOUT
PA3/TIM2_CH4/TIM5_CH4/TIM9_CH2/SAI1_FS_A/USART2_RX/OTG_HS_ULPI_D0/EVENTOUT
PA4/SPI1_NSS/I2S1_WS/SPI3_NSS/I2S3_WS/USART2_CK/OTG_HS_SOF/DCMI_HSYNC/EVENTOUT
PA5/TIM2_CH1/TIM2_ETR/TIM8_CH1N/SPI1_SCK/I2S1_CK/OTG_HS_ULPI_CK/EVENTOUT
PA6/TIM1_BKIN/TIM3_CH1/TIM8_BKIN/SPI1_MISO/I2S2_MCK/TIM13_CH1/DCMI_PIXCLK/EVENTOUT
PA7/TIM1_CH1N/TIM3_CH2/TIM8_CH1N/SPI1_MOSI/I2S1_SD/TIM14_CH1/RMC_SDNWE/EVENTOUT
PA8/MCO1/TIM1_CH1/I2C3_SCL/USART1_CK/OTG_FS_SOF/EVENTOUT
PA9/TIM1_CH2/I2C3_SMBA/SPI2_SCK/I2S2_CK/SAI1_SD_B/USART1_TX/DCMI_D0/EVENTOUT
PA10/TIM1_CH3/USART1_RX/OTG_FS_ID/DCMI_D1/EVENTOUT
PA11/TIM1_CH4/USART1_CTS/CAN1_RX/OTG_FS_DM/EVENTOUT
PA12/TIM1_ETR/USART1_RTS/SAI2_FS_B/CAN1_TX/OTG_FS_DP/EVENTOUT
PA13/JTMS-SWDIO/EVENTOUT
PA14/JTCK-SWCLK/EVENTOUT
PA15/JTDI/TIM2_CH1/TIM2_ETR/HDMI_CEC/SPI1_NSS/I2S1_WS/SPI3_NSS/I2S3_WS/UART4_RTS/EVENTOUT
PB0/TIM1_CH2N/TIM3_CH3/TIM8_CH2N/SPI3_MOSI/I2S3_SD/UART4_CTS/OTG_HS_ULPI_D1/SDIO_D1/EVENTOUT
PB1/TIM1_CH3N/TIM3_CH4/TIM8_CH3N/OTG_HS_ULPI_D2/SDIO_D2/EVENTOUT
PB2-BOOT1(PB2)/TIM2_CH4/SAI1_SD_A/SPI3_MOSI/I2S3_SD/QUADSPI_CLK/OTG_HS_ULPI_D4/SDIO_CK/EVENTOUT
PB3(JTDO/TRACESWO)/TIOGTRACESWO/TIM2_CH2/I2C2_SDA/SPI1_SCK/I2S1_CK/SPI3_SCK/I2S3_CK/EVENTOUT
PB4(NJTRST)/NJTRST/TIM3_CH1/I2C3_SDA/SPI1_MISO/SPI3_MISO/I2S2_NSS/I2S2_WS/EVENTOUT
PB5/TIM3_CH2/I2C1_SMBA/SPI1_MOSI/I2S1_SD/SPI3_MOSI/I2S3_SD/CAN2_RX/OTG_HS_ULPI_D7/FMC_SDCKE1/DCMI_D10/EVENTOUT
PB6/TIM4_CH1/HDMI_CEC/I2C1_SCL/USART1_TX/CAN2_TX/QUADSPI_BK1_NCS/FMC_SDNE1/DCMI_D5/EVENTOUT
PB7/TIM4_CH2/I2C1_SDA/USART1_RX/SPDIFRX_IN0/FMC_NL/DCMI_VSYNC/EVENTOUT
PB8/TIM4_CH3/TIM2_ETR/TIM4_CH3/TIM10_CH1/I2C1_SCL/CAN1_RX/SDIO_D4/DCMI_D6/EVENTOUT
PB9/TIM4_CH0/TIM4_CH4/TIM11_CH1/I2C1_SDA/SPI2_NSS/I2S2_WS/SAI1_FS_B/CAN1_TX/SDIO_D5/DCMI_D7/EVENTOUT
PB10/TIM2_CH3/I2C2_SCL/SPI2_SCK/I2S2_CK/SAI1_SCK_A/USART3_TX/OTG_HS_ULPI_D3/EVENTOUT
PB11/TIM2_CH4/I2C2_SDA/SPI2_NSS/I2S2_WS/SAI1_SCK_B/USART3_CK/CAN2_RX/OTG_HS_ULPI_D5/OTG_HS_ID/EVENTOUT
PB13/TIM1_CH1N/SPI2_SCK/I2S2_CK/USART3_CTS/CAN2_TX/OTG_HS_ULPI_D6/EVENTOUT
PB14/TIM1_CH2N/TIM8_CH2N/SPI2_MISO/I2S2ext3_RTS/TIM12_CH1/OTG_HS_DM/EVENTOUT
PB15/RTC_REFIN/TIM1_CH3N/TIM8_CH3N/SPI2_MOSI/I2S2_SD/TIM12_CH2/OTG_HS_DP/EVENTOUT
PC0/SAI1_MCLK_B/OTG_HS_ULPI_STP/FMC_SDNWE/EVENTOUT
PC1/SPI3_MOSI/I2S3_SD/SAI1_SD_A/SPI2_MOSI/I2S2_SD/EVENTOUT
PC2/SPI2_MISO/OTG_HS_ULPI_DIR/FMC_SDNE0/EVENTOUT
PC3/SPI2_MOSI/I2S2_SD/OTG_HS_ULPI_NXT/FMC_SDCKE0/EVENTOUT
PC4/I2S1_MCK/SPDIFRX_IN2/FMC_SDNE0/EVENTOUT
PC5/USART3_RX/SPDIFRX_IN3/FMC_SDCKE0/EVENTOUT
PC6/TIM3_CH1/TIM8_CH1/FMIPI2C1_SCL/I2S2_MCK/USART6_TX/SDIO_D6/DCMI_D0/EVENTOUT
PC7/TIM3_CH2/TIM8_CH2/FMIPI2C1_SDA/SPI2_SCK/I2S2_CK/I2S3_MCK/SPDIFRX_IN1/USART6_RX/SDIO_D7/DCMI_D1/EVENTOUT
PC8/TRACED0/TIM3_CH3/TIM8_CH3/UART5_RTS/USART6_CK/SDIO_D0/DCMI_D2/EVENTOUT
PC9/MCO2/TIM3_CH4/TIM8_CH4/I2C3_SDA/I2S_CKIN/UART5_CTS/QUADSPI_BK1_IO0/SDIO_D1/DCMI_D3/EVENTOUT
PC10/SPI3_SCK/I2S3_CK/USART3_TX/UART4_TX/QUADSPI_BK1_IO1/SDIO_D2/DCMI_D8/EVENTOUT
PC11/SPI3_MISO/USART3_RX/UART4_RX/QUADSPI_BK2_NCS/SDIO_D3/DCMI_D4/EVENTOUT
PC12/I2C2_SDA/SPI3_MOSI/I2S3_SD/USART3_CK/UART5_TX/SDIO_CK/DCMI_D9/EVENTOUT
PC13/EVENTOUT
PC14-OSC32_IN(PC14)/EVENTOUT
PC15-OSC32_OUT(PC15)/EVENTOUT
PD2/TIM3_ETR/UART5_RX/SDIO_CMD/DCMI_D11/EVENTOUT
PH0-OSC_IN/EVENTOUT
PH1-OSC_OUT/EVENTOUT
NRST
VSSA/VREF-
VSS
VDD
VDDA/VREF+
VBAT
VCAP_1
C1
C4
C7
C8
C5
C9
4.7u
C3
C2
1u
BAT_SENSN
DRB_CSN
KICKER(D0)
SD1614
UART3_RX
UART6_TX
UART6_RX
TIM5_CH1
SPWR
MB_SUB1
MB_SUB2
UART5_TX
GOAL_SWITCH
IR1
IR2
UART5_RX
HSE-IN
HSE-OUT
NRST
S1
C10
100n
C12
15p
C11
15p
XTAL1
OSC_IN
OSC_OUT
R1
10k
R2
4.7k
R3
4.7k
R20
1k
C30
100n
DRB_CSN
DRB_CS
D3R7
SS34SMFL
R12
300
Q1
2SC2712-YLF
SPWR
POLARITY_PAD_(+)
POLARITY_PAD_(-)
DUMMY_PAD_2
DUMMY_PAD_1
U2
UGC7525AN4
R16
R17
R18
R19
C26
C27
C28
C29
100n
CURRENT_SENS_1
CURRENT_SENS_2
CURRENT_SENS_3
CURRENT_SENS_4
CURRENT_SENS_1IN
CURRENT_SENS_2IN
CURRENT_SENS_3IN
CURRENT_SENS_4IN
C20
100n
NRST2
R15
10k
C21
C22
C24
C23
100n
10u
4.7u
C25
1u
BOOT0
NRST
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB3
PB4
PB5
PB6
PB7
PF0/OSC_IN
PF1/OSC_OUT
VDD_1
VDD_2
VDD/VREF+
VSS_1
VSS_2
STM32F303K8T6
J4
J5
J6
VCC
TRIG
ECHO
GND
RCW-0001